

Beyond predictive benchmarks: infrastructure for hypothesis-driven cognitive testing of large language models

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Abstract

Large language models are increasingly studied as psychological subjects, but the communities doing so apply incompatible standards of evidence: NeuroAI evaluates a model by how well it predicts responses to naturalistic stimuli, while psychology evaluates it by how well it accounts for the effect of manipulating a controlled stimulus to test a falsifiable, mechanism-level hypothesis. Running the second kind of study at the scale quantitative psychology expects currently has no dedicated infrastructure: prompt construction, response parsing, memory management, persona conditioning, and feedback injection are hand-built for every new task and model family. We present **psychscanner**, an open-source framework in which a researcher authors a controlled cognitive or survey task once, as a declarative task card, and runs it across memory architectures, persona manipulations, feedback regimes, agent architectures, and model families without rewriting the underlying mechanics. We describe the framework’s architecture and report a six-module demonstration suite exercising its feature surface on real model calls. Two modules have complete data. A three-armed bandit task run through three agent architectures found none produced explore-then-exploit behaviour, and that the dominant effect was instruction-following collapse under structured feedback rather than an architecture difference. A 480-trial factorial replication of a classic imagery paradigm found no measurable effect of persona conditioning on vividness ratings, but that single-turn prompting returned an identical rating on all 160 of its trials — a response-degeneracy artifact invisible to any aggregate score. Because every model backend implements one interface, the same task cards additionally capture residual-stream activations, which we validate by reproducing the Othello-GPT board-state probing result on a toy transformer trained for the purpose. In both completed behavioural demonstrations the informative result sat one level below the headline metric and became visible only because the design varied one factor at a time.

1 Introduction

Large language models are increasingly treated as objects of psychological inquiry in their own right (??) — evaluated for their memory, personality self-report, imagery, and reasoning much as a human participant would be. But the research communities doing this evaluation do not share a common methodology, and the resulting disagreements are not merely about findings; they are about what counts as evidence in the first place. NeuroAI typically evaluates a model by how well it predicts behavioural and neural responses to naturalistic stimuli, treating out-of-sample predictive variance on broad benchmarks as the mark of success and alignment with biological computation as the underlying goal (?). Psychology, by contrast, evaluates a model by how well it accounts for the effect of systematically manipulating a controlled, artificial stimulus — changing one factor at a time

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to test a specific, falsifiable hypothesis about a specific mechanism (?). These are not interchangeable standards: a model can predict naturalistic behavioural data well while getting the underlying mechanism wrong, and a model narrowly tuned to a controlled manipulation may generalize poorly to naturalistic settings. Recent work has made this tension explicit as a live methodological disagreement rather than a settled question, with researchers reaching opposite conclusions about how much NeuroAI’s predictive successes actually tell us about biological intelligence.¹

This disagreement is not abstract, and it is not confined to vision models. In a June 2026 preprint, Bowers and colleagues make the general critique concrete against a large language model: they argue that Centaur (?), fine-tuned to predict human trial-by-trial responses across a large database of psychology experiments, is not subject to the cognitive constraints it is meant to model, that its behaviour is better characterised as role-play than as a mechanistic account, and more generally that predictive power is being mistaken for explanatory power (?). The GAC’s own comment thread reproduces the disagreement in real time and narrows it to something closer to a resolvable empirical question. Responding to Bowers’s opening challenge — what have we learned about mind and brain from correlational studies with ANNs? — Michael Tarr argues that the relevant distinction is not whether a study is correlational but whether a model comparison is *theoretically diagnostic*, and that NeuroAI already runs manipulation-based science, just inside the model’s architecture and training rather than on its input, pointing to convergent food-selectivity findings in ventral visual cortex across independent studies using different ANN-guided methods as an example of what this approach has taught us. Bowers’s reply narrows the disagreement to a concrete result: manipulating the images behind a widely used behavioural benchmark collapses a leading brain-similarity metric’s prediction accuracy from near 100% to under 20%, which he reads as evidence that predictive success is often driven by shortcuts rather than shared mechanism (?). Both positions concede the same limit. Tarr’s own defence of NeuroAI treats manipulating a model’s assumptions as the diagnostic move; and Bowers’s closing position in the thread is explicitly not to abandon ANNs but to “complement this new approach with old fashioned methods” (?) — systematic experimental manipulation, at scale, which is exactly the missing infrastructure this paper builds for large language models. Psychology’s own history is a warning against settling for the alternative: more than five decades before this GAC, Newell argued that testing isolated hypotheses about the mind one at a time, without shared infrastructure for accumulating results across many controlled studies into a working theory, could not by itself add up to psychology of the kind that explains cognition rather than merely describing pieces of it (?). psychscanner is an attempt to give hypothesis-driven, manipulation-based testing of LLMs that shared, reusable infrastructure, at the exact moment this methodological choice is being litigated in public rather than assumed.

Applying psychology’s side of this methodology to LLMs at any scale — systematically manipulating one factor (memory architecture, persona, feedback, stimulus modality) across many models, many replications, and many controlled task variants, the way a human cognitive psychology lab would run a study — has no dedicated infrastructure. A researcher wanting to ask a falsifiable, mechanism-level question (e.g. does giving a model conversational memory change how consistently

¹See the 2026 Cognitive Computational Neuroscience (CCN) Generative Adversarial Collaboration organized by Jeffrey Bowers and colleagues, “Is NeuroAI adopting the right methods and theoretical frameworks to advance our understanding of mind and brain?” (?), whose kickoff workshop set Milton Montero (current NeuroAI methods are problematic) against Martin Schrimpf (current methods are appropriate), with Jenelle Feather and Anya Ivanova arguing a middle position. The GAC’s own proposal frames the disagreement across five methodological axes: the goal and scope of modelling, whether engineering success is a valid starting point, whether naturalistic or controlled stimuli are the right evaluation target, how a model should be evaluated (predictive variance vs. manipulation-driven effect sizes), and the role of falsification.

it reports a stable personality across a survey, the way it would for a human respondent?) has to hand-write the experiment scaffolding — prompt construction, response parsing, memory/state management, persona conditioning, feedback injection — from scratch for every new task and every new model family, which is exactly the kind of ad hoc, one-off tooling that makes systematic, multi-condition, multi-model replication impractical.

We introduce **psychscanner**, an open-source framework that closes this infrastructure gap: it lets a researcher author a controlled cognitive or survey task once, as a declarative task card, and run it across memory conditions, persona manipulations, feedback regimes, and model families without rewriting any of the underlying mechanics. This is not a benchmark and not a claim that models are or are not human-like — it is the missing methodological toolkit for asking the kind of manipulation-based, falsifiable question that psychology’s half of the NeuroAI/psychology divide calls for, applied to LLMs specifically, at the scale quantitative psychology expects.

2 Background and related work

Behavioural machine psychology. Framing LLMs as experimental subjects has produced evidence of human-like cognitive biases and persona-contingent behaviour shifts (??), and the methodology for doing so is now being formalized explicitly (?). Benchmarks such as PsychoBench (?) and task ontologies such as the Cognitive Atlas (?) give this approach structure and coverage targets. What they share is a limit rather than a flaw: they characterize what a model does, and none is designed to ask why a given response pattern occurred. Two studies run on earlier versions of the framework described here sit squarely inside that limit. The first found that LLM source-monitoring accuracy — distinguishing self-generated from externally-provided content — degrades under conversational delay in a pattern that tracks active rather than aggregate parameter count, producing feedback-induced failure modes invisible to existing benchmarks (?). The second found that LLM imagination-vividness ratings produce a degenerate, single-cluster network topology (median ARI ≈ 0) where three geographically distinct human populations show robust, replicable community structure ($r = 0.31\text{--}0.93$) (?). Both are behavioural results whose mechanism is unknown.

Neuroscience-inspired interpretability. The methods that could answer the second question have direct neuroscience analogues, which is what makes the pairing natural rather than opportunistic. Linear probing for task-relevant internal state is methodologically the same operation as neural decoding from recorded population activity; its most visible language-model instance is the Othello-GPT result that a transformer trained only on legal-move sequences develops a linearly-decodable internal board representation (?). Causal tracing and rank-one model editing (ROME) localize and manipulate where a factual association is stored (?), functionally analogous to lesion and reversible-inactivation methodology in systems neuroscience. Contrast-Consistent Search (CCS) recovers a linear “truth direction” from unlabeled activations by enforcing logical consistency between a statement and its negation (?). Persona vectors (?) apply the same probing and steering toolkit to character traits such as sycophancy and hallucination propensity, extracted from natural-language trait descriptions and monitored or controlled at inference and fine-tuning time — activation steering being the analogue of stimulation along an identified pathway. psychscanner’s interpretability backend (§??) is designed to run all four method families through one experimental harness, on top of the same persona and task-card system already validated for behavioural work. We regard putting these methods inside a reusable experimental framework, rather than leaving them as one-off analysis scripts, as a contribution in its own right: it makes the mechanistic toolkit available on the same platform that produced the behavioural questions in the first place.

Why behavioural and mechanistic testing belong in one harness. Interpretability research has itself recently moved toward the conclusion that motivates keeping these two kinds of testing together rather than treating the mechanistic side as a self-sufficient replacement for behavioural measurement. DeepMind’s mechanistic interpretability team has argued for grounding the field in a pragmatic proxy task and starting from the simplest method that could work, citing disappointment with the progress delivered by ambitious reverse-engineering of model internals (?). A concurrent clinical-triage study makes the resulting gap concrete: linear probes recovered hazardous cases from internal activations at 98.2% AUROC, yet steering on those same directions corrected only 20% of the errors the model’s output actually made, and disrupted 53% of the cases it was already getting right (?) — near-perfect internal representations that none of four steering and probing methods could reliably turn into corrected behaviour. Read together, these results say that internal state is a complement to behavioural measurement, not a substitute for it, which is why psychscanner’s interpretability backend sits behind the same model-provider seam as its behavioural task cards rather than as a separate analysis pipeline. Even inside interpretability the organizing vocabulary is borrowed from cognitive psychology: a recent survey of attention-head function proposes a four-stage framework modelled explicitly on the human thought process — knowledge recalling, in-context identification, latent reasoning, and expression preparation — to categorize what different attention heads do (?). That a psychological stage model is doing organizing work inside mechanistic interpretability is itself a data point for the pairing this paper argues for: the two literatures are already asking about the same stages of processing, from different kinds of data.

3 Framework overview

Figure ?? gives the component structure of the released package (v0.3.0). Every run traverses the same spine. The researcher’s task card and persona files are resolved into an `ExpCard`, which is expanded into one system message per design cell and one message per trial; `TaskRunner` executes those trials against a configured agent; and the results are validated and written to disk as one record per participant, exported on demand to a tidy one-row-per-trial table. Around that spine sit four seams the researcher is expected to extend — response parsers, model providers, agent architectures, and trial-level feedback — and it is the model-provider seam that carries the interpretability backends (§??). The design principle throughout is that anything a psychologist would want to manipulate as an experimental factor should be a declaration on the card, not a code path; the subsections below describe each part in those terms.

3.1 Task cards

The unit a researcher actually authors is a *task card*: a plain JSON document (or an equivalent inline Python dict) declaring an experiment’s instructions, its named contexts, and its trials, with no code required for the common case. Each trial carries a unique `trcode` and a stimulus — plain text, a structured dict, or a list of multimodal content blocks (image/audio/text) for vision-language tasks. The text before the first underscore in a trial’s `trcode` is matched against the card’s `contexts_id` list to derive a `context_item`: the full-text description of whichever condition that trial belongs to. This is computed once at prompt-build time and attached to every trial automatically, for both prompting (when `context_present` is set) and for output records — an author writes the context-to-condition mapping once, rather than repeating context text on every trial.

A single `chain_type` field, independent of which memory mode (stateless `SingleTurn` or continuous `Convo`) is active, controls how trials share conversational state: `item` sends each stimulus as an independent model call; `trial` groups the multiple stimuli under one `trcode` into a single

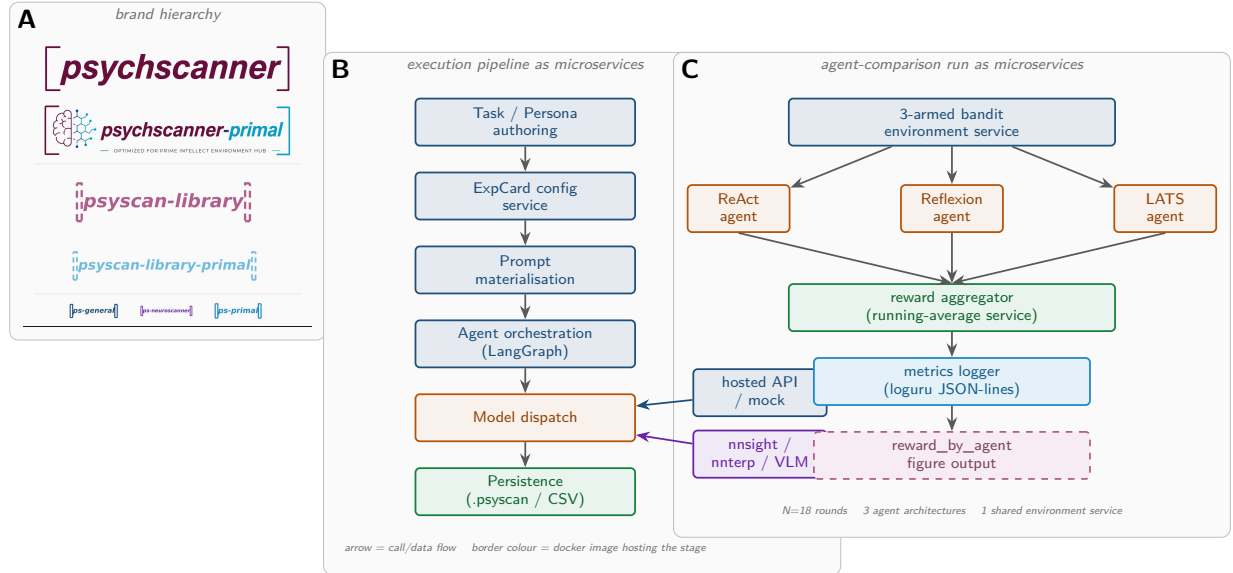


Figure 1: The PsychScanner project as a microservices system. (A) Brand hierarchy: the two framework repositories (psychscanner, psychscanner-primal), their companion vetted task/experiment-card libraries (psyscan-library, psyscan-library-primal), and the three published Docker images (ps-general, ps-neuroscanner, ps-primal). (B) The single-run execution pipeline, redrawn as a service pipeline; box border colour marks which Docker image hosts each stage. (C) The multi-agent bandit comparison behind Fig. ??, redrawn the same way: one shared environment service, three interchangeable agent services, and a reward-aggregation/logging path terminating in that figure.

conversational thread (e.g. a study phase immediately followed by its matching test phase); **task** places every trial for the whole run on one continuous thread. This one parameter is what lets a single trial-execution engine implement single-turn, trial-chain, and episodic-chain designs — the three trial structures demonstrated across Demos 2 and 3 (§??–§??) and the extended factorial suite in Supplementary Information Sections S1–S2 — without separate code paths for each. Structured response parsing (a Pydantic schema per trial or per card, resolved by name or passed directly), a per-trial subset of the card’s bound tools, and trial-level feedback injection (fed back into the next trial’s prompt) are declared the same way, as optional fields on the same card, so a researcher varying these as experimental conditions edits one card rather than one code path per condition.

3.2 Persona conditioning and memory conditions

Persona conditioning is authored as one or more JSON files, each holding a list of persona statements. At prompt-build time the framework takes the Cartesian product across those files and emits one system message per resulting cell, so a design crossing two persona files with two levels each yields four conditions without the researcher enumerating them by hand; this is the machinery the WEIRD / non-WEIRD manipulations in §?? and Supplementary Information S1 are built from.

Memory is configured independently of persona. **SingleTurn** compiles the execution graph without a conversation checkpoint; **Convo** compiles it with one. Windowing and summarization are then orthogonal parameters rather than additional modes: **memory_k** bounds how much history is retained, and **summary_k** folds the overflow into a rolling summary prepended to the system message.

Four commonly-reported memory architectures — stateless, unbounded conversation, windowed conversation, and windowed rolling summary — are therefore two parameter settings rather than four implementations, which is what makes memory crossable with persona and feedback inside a single factorial design (§??).

3.3 Agent architectures and feedback

The component that executes a trial is swapped through one `custom_agent=` argument, against a small protocol the framework only ever calls two members of. Alongside a bare single-call baseline, the package ships ReAct, supervisor, planner-executor, and three reflection variants (basic reflection, Reflexion, and LATS). Because they sit behind the same protocol, agent architecture becomes a manipulable factor in an otherwise identical design — which is exactly what §?? exploits.

Trial-level feedback is supplied by subclassing a single abstract class with one method, which receives the trial and the parsed response and returns a string; the framework folds that string into the *next* trial’s prompt, handling the multimodal case as well as plain text. The package deliberately ships no concrete feedback subclasses: reward, correctness, and value-tracking feedback are experiment-local definitions, because what counts as feedback is part of the experimental design rather than the infrastructure.

3.4 Model providers and the interpretability backend

Every model backend is reached through a single dispatch function and implements the same `LangChain BaseChatModel` interface, so which model executes a trial is a field on the `ExpCard` rather than a code path. Hosted APIs (OpenAI, Anthropic, Groq, Ollama, OpenRouter), a deterministic mock model for offline testing, and three local backends built on `mnsight` are all interchangeable at that one seam.

The local backends are what make mechanistic work possible without a parallel pipeline. `ChatNNsightModel` wraps a local Hugging Face model and captures activations from a researcher-specified list of dotted module paths; `ChatNNterpModel` wraps a standardized transformer interface and addresses activations by layer index crossed with kind (layer, attention, or MLP output) instead, which is the more convenient addressing scheme when the same probe is to be run across model families; and a vision-language variant extends the same capture to multimodal prompts. Each call runs a cached forward pass over the prompt and writes the captured per-module residual-stream activations to disk, referencing the file from the returned message rather than embedding tensors in it — mirroring how the framework externalizes multimodal stimuli.

Two consequences are worth stating explicitly. First, because this sits behind the ordinary model-provider interface, every existing mechanism — persona conditioning, memory modes, task cards, feedback hooks — applies unchanged: a persona-contrast experiment that would previously have yielded only behavioural response differences now also yields paired activation captures, with no separate extraction pipeline to write. Second, capture is over the prompt’s forward pass and not per generated token, which bounds what can legitimately be asked of the resulting tensors and is a real limitation of the current implementation rather than a configuration detail.

3.5 Outputs and reproducibility

Each participant’s trials are validated against a nested Pydantic schema and written as a single record; multimodal stimuli are externalized to a content-addressed media store, so a run’s outputs contain hashes rather than repeated base64 blobs. A separate export step flattens those records into a tidy one-row-per-trial table carrying the condition labels (`memory`, `chain_type`, `context_item`,

`trcode`, and the parsed response fields) alongside the raw response, so the analysis step is an ordinary dataframe operation. A session tunnel logs checkpoints throughout, so an interrupted multi-participant run resumes at the last completed simulation rather than restarting — a practical requirement for the multi-hundred-trial, rate-limited runs these designs produce. The full configuration round-trips to JSON, so a run is describable by one file plus the task card.

3.6 Comparison with related frameworks

Table ?? places psychscanner against six frameworks a reader might otherwise reach for. Three are the accuracy-first evaluation harnesses already discussed in §??: the EleutherAI `lm-evaluation-harness` (?), BIG-bench (?), and the AISI Inspect framework (?). The other two, AutoRA and SweetBean, come from a closely related but distinct tradition: Sebastian Musslick’s lab builds tooling for closed-loop behavioural-science automation rather than LLM-as-participant cognitive testing specifically. AutoRA automates the scientific-discovery loop itself — an experimentalist proposes conditions, an experiment runner collects data (from human participants via Prolific, a synthetic/equation experiment, or another runner), and a theorist fits a model, with the cycle repeating (?); it has no concept of a persona, a conversational LLM participant, or a task card, and is not a fit for the same job psychscanner does. SweetBean is the closer comparison: a declarative Python DSL that compiles one experiment specification into either a jsPsych timeline for human participants or a text-based prompt sequence for an LLM “synthetic participant” (?), and explicitly interoperates with AutoRA for closed-loop deployment. Ratings below for AutoRA and SweetBean are drawn from their public documentation, JOSS papers, and source (github.com/AutoResearch/{autora,sweetbean}, accessed 2026-08-13) using the same three-level scale as the original comparison: SweetBean ships a first-class `Feedback` stimulus for trial-by-trial correctness feedback and reinforcement, and its response specifications carry explicit `correct_*` fields, so it scores accuracy natively — stronger than psychscanner’s own “Partial” rating on that row. Persona conditioning in SweetBean is a single `demographic(age, gender)` prompt template rather than an arbitrary persona-statement system; we rate that Partial. Neither AutoRA nor SweetBean documents a dedicated crash-safe checkpoint/resume subsystem for long API sweeps comparable to `SessionTunnel`.

Table 1: Feature comparison of LLM/behavioural evaluation frameworks. ✓ = first-class support; “Partial” = limited or requires a workaround; ✗ = not supported. psychscanner, Eval Harness, Inspect, and BIG-bench ratings are based on public documentation and source code (???). AutoRA and SweetBean ratings are based on their JOSS papers and public source (??), accessed 2026-08-13.

Feature	psychscanner	Eval Harness	Inspect	BIG-bench	AutoRA	SweetBean
Persona conditioning	✓	Partial	✓	Partial	✗	Partial
Within-session memory modes	✓	Partial	✓	Partial	✗	Partial
Trial-level feedback	✓	✗	✓	✗	✗	✓
Session resume / checkpoints	✓	Partial	Partial	✗	Partial	✗
Structured response schemas	✓	Partial	Partial	✗	✗	Partial
Portable experiment config	✓	✗	✗	✓	Partial	Partial
JSON task specification	✓	✓	✓	✓	✗	Partial
Accuracy benchmarking	Partial	✓	✓	✓	✗	✓

The comparison sharpens rather than flattens psychscanner’s positioning. Against the accuracy harnesses, its primary differentiators remain structured typed response schemas enforced at the model level, memory modes mapped directly onto between-/within-subjects designs, per-trial adaptive feedback hooks, and portable experiment serialization (`save_expcard/load_expcard`). Against

AutoRA, the two frameworks are complementary rather than competing: AutoRA’s experimentalist/theorist loop is a natural consumer of psychscanner-style task cards as an `experiment_runner`, a direction we flag as future work rather than a present capability. Against SweetBean, the comparison is closer and genuinely double-edged — SweetBean’s native accuracy scoring and first-class feedback stimulus are real strengths psychscanner lacks in equal form, while psychscanner’s persona-file system, three explicit memory architectures, and typed per-task response parsing go further than SweetBean’s single-dimension demographic prompting and general-purpose response specs built for jsPsych’s input types rather than free-form model output.

4 Results

Six demonstration modules (`examples/demonstration_suite/01_reward_task` through `06_advanced_demonstration`) exercise psychscanner’s feature surface end-to-end on real model calls; each is summarized below from its own `reports/report.md`, generated directly from the run that produced it (no numbers in this section are estimated or reused across demos). Table ?? summarizes the design and current data-collection status of all six before the per-demo detail that follows.

Table 2: Demonstration-suite design overview. Status reflects data collection as of this draft, not design completeness — all six task cards, memory/persona/feedback conditions, and analysis scripts are implemented regardless of how much data has been collected against them.

Demo	Task	Key conditions	Model(s)	Data status
01	3-armed bandit	3 agent architectures	llama-3.1-8b-instant (Groq)	Complete, analyzed
02	PAL-50	single-turn / trial-chain / episodic-chain × feedback	TBD	Not started
03	BFI-44 survey	2 persona × 4 memory	llama-3.3-70b-versatile (Groq)	1/8 cells
04	Shape-naming VQA	2 memory × 2 feedback	gemma-4-26b-a4b-it:free (OpenRouter)	1/4 conditions
05	Persona-vector probing	trait extraction/steering	gpt2/gpt2-medium	Not started
06	Othello/ROME/CCS	3 toy reproductions	toy transformer (CPU)	1/3 (Othello only, §??)

4.1 Demo 01: Agent-architecture swapping on a reward-learning task

A 3-armed bandit task (hidden reward means 2.0/6.0/4.0, value-tracking feedback fed back each round) was run through three structurally different `psychscanner.agents` implementations — a bare single-call baseline (`CustomAgent`), a generate/reflect self-critique loop, and a plan/execute/validate loop — plugged into the identical `ExpCard/ScannerModel/FeedbackBase` pipeline via a single `custom_agent=` argument, on `llama-3.1-8b-instant` (Groq), 6 rounds × 1 participant per agent (18 scored rounds total, real API calls, no mocking).

None of the three architectures produced textbook explore-then-exploit behavior: the baseline accumulated 16.14 total reward (mean 2.69/round, 0% optimal-arm picks), self-reflection 14.14 (mean 2.36/round, 16.7% optimal, the only agent to reach the optimal arm at all), and plan/execute/validate 12.14 (mean 2.02/round, 0% optimal; Table ??). All three under-selected the optimal arm (true mean 6.0) in favor of the worst arm (true mean 2.0), and none approached the optimal arm’s mean over the course of the run (Fig. ??).

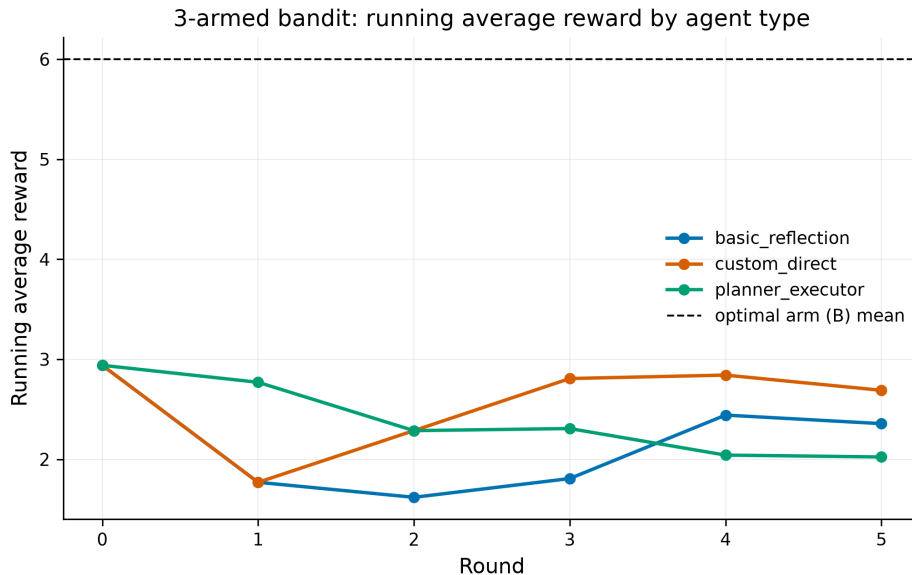


Figure 3: Running average reward by agent architecture across the 3-armed bandit task (Demo 01). The optimal arm’s true mean (6.0) is shown as a dashed reference line.

Table 3: Demo 01 per-agent summary. $N = 18$ rounds total (6 rounds $\times$ 3 architectures), demonstration-scale, not a powered study.

Agent architecture	Total reward	Mean reward/round	% optimal-arm picks
Baseline (single-call)	16.14	2.69	0%
Self-reflection	14.14	2.36	16.7%
Plan/execute/validate	12.14	2.02	0%

More important than the reward gap itself is a confound the report flags plainly: from round 2 onward, once JSON-formatted trial feedback entered the prompt, the 8B-parameter model mostly stopped emitting the requested single-letter choice and instead narrated the JSON structure in prose, across all three agent types. A regex fallback still extracted a letter from these responses, so trials scored, but the recorded “choice” is partly confounded with incidental letter-mentions rather than deliberate decisions. The dominant effect observed here is instruction-following collapse under a structured-feedback format on a small model, not an architecture effect — worth stating plainly rather than reading the per-agent reward gap as a clean architecture comparison. See [examples/demonstration_suite/01_reward_task/reports/report.md](#) for full detail, including the caveat that this is an $N=18$ -round demonstration run, not a powered study.

4.2 Demo 02: Paired-associate learning across memory/feedback conditions

A PAL-50 task (50 word pairs, study phase then test phase) exercises five of psychscanner’s trial-structure and feedback primitives in one design: single-turn (independent study/test trials), trial-chain (paired encoding+test within one thread), and episodic-chain (all trials in one continuous conversation) memory architectures, crossed with correct/incorrect feedback and reward feedback. A more elaborate factorial replication of the same paradigm — three memory architectures $\times$ five semantic-similarity levels $\times$ feedback presence — is reported in Supplementary Information S2, with

the same PAL-50 task and scoring logic. **[PLACEHOLDER: recall-accuracy results by condition and similarity level, pending simulation/run_simulation.py + analysis/analyze.py output — no simulation data collected for this core demo as of this draft.]**

4.3 Demo 03: Personality survey across persona and memory conditions

A BFI-44 (44-item, OCEAN-trait) survey is run across 2 persona conditions (`weird`, `non_weird_jung`) $\times$ 4 memory conditions — stateless (`SingleTurn`), unbounded conversation (`Convo`), windowed conversation (`memory_k=8`), and windowed rolling summary (`memory_k=8`, `summary_k=4`) — via real model calls (Groq, `llama-3.3-70b-versatile`). This demonstrates persona conditioning together with all four memory-window configurations psychscanner supports in one factorial design. The companion factorial persona/survey suite in Supplementary Information S1 covers the same persona $\times$ memory space plus a completed VVIQ-16 imagery arm (S1.1); see §?? for that result, including a response-degeneracy finding under single-turn conditions that is directly relevant to interpreting any BFI-44 self-report scores from this demo once analyzed. **[PLACEHOLDER: per-trait persona/memory effect sizes, pending full 8-cell data collection — 1 of 8 cells (`weird/stateless`, `gpt-oss-120b`) collected as of this draft.]**

4.4 Demo 04: Vision-language task with reward/accuracy feedback

A shape-naming VQA task (6 procedurally-drawn colour+shape images, each shown twice per participant across two blocks) crosses 2 memory conditions (`SingleTurn` vs. `Convo` with rolling summary, `memory_k=6`, `summary_k=3`) with 2 feedback conditions — demo-local `FeedbackBase` subclasses defined in this demo’s own simulation script, not framework classes — (`RewardFeedback`: ± 1 score only; `AccuracyFeedback`: explicit correct/incorrect plus the right answer), via real vision-model calls (OpenRouter, `google/gemma-4-26b-a4b-it:free`). Repeat exposure across the two blocks lets the analysis separate one-shot accuracy from learning-from-feedback effects. Two more elaborate multimodal factorial designs — a Behavioral Profiling Task (OCR-VLM vs. non-OCR-VLM $\times$ feedback) and a VISER visual-structure replication — are reported in Supplementary Information S3–S4. **[PLACEHOLDER: accuracy by memory $\times$ feedback $\times$ block, pending full 4-condition data collection — 1 of 4 conditions (`singleturn_reward`) collected as of this draft.]**

4.5 Demo 05: Persona-vector extraction, probing, and steering

Following the Persona Vectors approach (?), this demo extracts residual-stream activation directions for specific traits (e.g. sycophancy, hallucination) from psychscanner’s existing persona-conditioned trials via the `nnsight/nnterp` backend, fits a linear probe for each trait direction, and steers generation by adding or subtracting the extracted vector — mapping an interpretability technique directly onto psychscanner’s persona-conditioning infrastructure rather than requiring separate activation-collection tooling. This demo is also core content of the companion NeurIPS NeuroAI workshop submission (`papers/neurips-neuroai/`). **[PLACEHOLDER: probe accuracy and steering effect sizes per trait, pending simulation data — not yet collected as of this draft.]**

4.6 Demo 06: Othello-GPT / ROME / CCS toy reproductions

A capstone interpretability demo reproducing three classic mechanistic-interpretability studies on top of the infrastructure built for Demo 05: linear probing of residual-stream activations for an internal board-state world model (Othello-GPT; ?), causal tracing and rank-one weight editing to

locate and edit factual associations (ROME; ?), and unsupervised discovery of a “truth direction” via contrast-consistent search (CCS; ?). Also core content of the companion NeurIPS NeuroAI workshop submission. One of the three has been run.

Othello-GPT. A 4-layer, CPU-tractable toy transformer ($d_{\text{model}} = 64$, 4 attention heads) was trained for 12 epochs on 4,000 games represented purely as legal-move sequences — board state was never part of the training signal — and then linearly probed for board-state decodability at each residual-stream layer on 400 held-out probe games. Legality checking on the trained model (728 boundary steps) placed a mean probability mass of 0.998 on legal moves against a random baseline of 0.75, confirming the model had learned the task before its internals were probed.

Board-state decodability sits near the shuffled-label baseline at layer 0 (0.423 vs. 0.140) and jumps above 0.93 from layer 1 onward (Table ??). That is the same qualitative layer-wise pattern the original study reported (?): early layers encode surface features of the move sequence, and task-structured internal state becomes linearly accessible only once the network has had a layer to compose it. The pattern itself is a mechanistic-interpretability commonplace; what this run establishes is infrastructural, namely that the whole path — train, capture activations through the ordinary model-provider interface, probe, report — works on a model we trained ourselves rather than a shared checkpoint, through a framework originally built for behavioural persona experiments.

Table 4: Demo 06, Othello-GPT substitute: linear-probe accuracy for an internal board-state representation, by residual-stream layer. The shuffled-label baseline is a probe of identical capacity fitted at the same layer to shuffled labels, which controls for probe capacity alone recovering structure.

Layer	Probe accuracy	Shuffled-label baseline	Above baseline
0	0.423	0.140	+0.283
1	0.941	0.148	+0.793
2	0.953	0.149	+0.804
3	0.931	0.141	+0.790

ROME and CCS. [PLACEHOLDER: neither has been run as of this draft. The ROME run script exists but has never been executed and has produced no output data of any kind; the CCS sub-experiment has not progressed past empty scaffold directories. Any claim that this backend is validated as a general-purpose probing platform is currently supported by one of the three planned reproductions, not all three.]

5 Discussion

The two demonstrations with complete, analyzed data share a structure worth naming explicitly, not as two isolated findings but as one pattern: in both, the informative result sat one level below the headline metric, and only became visible because the design varied one factor while holding the others fixed.

In the VVIQ-16 replication, the headline number — a mean vividness rating around 3.7, pooled across conditions — is unremarkable and would pass as a plausible imagery-report score under almost any evaluation that stopped there. Decomposed by memory architecture, it dissolves: the persona-conditioned single-turn condition returned the identical rating (4) on every one of 160 trials, which is not a vividness judgment but a response-degeneracy artifact, while persona conditioning itself, the factor the experiment was nominally designed to probe, showed no measurable effect on

the mean at all (WEIRD 3.69 ± 0.88 vs. non-WEIRD 3.70 ± 0.89). Neither factor alone explains the collapse: a persona-free replication of the same task and model under single-turn prompting shows concentrated but non-degenerate ratings ($M = 3.81$, $SD = 0.66$), broadening once conversational memory is added ($M = 3.73$, $SD = 0.99$) (?) — variance changes gradually with memory alone, and collapses completely only once persona conditioning is layered on top of the no-memory constraint. That interaction, not either manipulated factor in isolation, is the actual finding, and it is invisible to any evaluation that varies memory or persona but not both together. In the bandit demonstration, the headline comparison — which agent architecture accumulates more reward — is a clean story only if you don’t notice that all three architectures, not the weakest one, stopped emitting the requested single-letter choice once structured JSON feedback entered the prompt. Read either result at the aggregate level and it looks like ordinary noise or a modest effect size. Decomposed by the manipulated factor, both are actually about model behaviour breaking down under a specific condition (single-turn prompting; structured feedback), a distinction with no analogue in an evaluation regime built around predicting held-out naturalistic responses rather than manipulating controlled ones.

This is also the honest cross-cutting caveat across the wider demonstration suite, not just these two: where a demo’s data collection is underway (Demo 03: one of eight cells; Demo 04: one of four conditions), the visible pattern so far is the same kind of small-model instruction-following fragility that Demo 01 made explicit, not yet a clean condition or architecture effect. Whether that fragility is specific to the models tested here or a more general property of how small, cheaply-run models respond to the kind of structured, feedback-heavy prompting that controlled psychological designs require is itself an empirical question the framework is built to answer, once the remaining demonstrations and the extended factorial suite in Supplementary Information Sections S1–S4 finish collecting data: rerun each design’s smallest-scale condition across a range of model sizes, with model scale as an explicit manipulated factor rather than an incidental property of which model happened to be cheap to run, and ask whether the fragility pattern has a capacity threshold.

The Othello-GPT reproduction (§??) belongs to a different category and should not be read as a third instance of that pattern. It is not a finding about language-model psychology at all; it is a validation that the mechanistic half of the framework does what the architecture in §?? claims. That distinction matters for how much weight it can carry. What the reproduction establishes is that a model trained inside this pipeline, with activations captured through the ordinary model-provider interface, supports a probe that recovers a known layer-wise signature — so probing, and by construction the causal-tracing and steering methods that use the same captures, are available on the same platform as the behavioural task cards. What it does not establish is anything about the divergences in §??. Neither the reality-monitoring degradation (?) nor the imagination-network collapse (?) has been probed mechanistically here, and neither has the response-degeneracy effect this paper’s own VVIQ-16 arm reports. That is a gap rather than a hedge, and it names the next experiment precisely: run the reality-monitoring and VVIQ-16 task cards through the activation-capturing backend and ask whether the constant-rating pattern under single-turn prompting corresponds to reduced representational variance in the residual stream at the layers these probes already target, or whether it is a decoding-time artifact with no mechanistic signature at all. Because the backend sits behind the same interface as every other model provider, that is a configuration change to an existing task card rather than a new study — which is, in the end, the claim this paper is making about the framework.

Limitations. Two of six core demonstrations and one of four extended factorial sub-experiments (VVIQ-16) have complete, analyzed behavioural data as of this draft; the rest range from fully designed but unstarted (Demo 02; extended S2–S4) to partially collected (Demos 03–04). Both completed behavioural results are demonstration-scale (16–18 rounds; 480 trials), not powered studies.

The interpretability results rest on a single probe run per layer of one toy transformer, and on one of the three planned reproductions: ROME and CCS have not been run, and Demo 05’s persona-vector extraction and steering has no collected data. Activation capture is over the prompt’s forward pass rather than per generated token, which limits the questions the current captures can answer. The framework’s contribution — making this class of manipulation-based, multi-condition, multi-model evaluation practical to run at all — is demonstrated; a comprehensive empirical picture of what that evaluation reveals about large language model cognition is not yet complete, and this paper does not claim otherwise.

Supplementary Information: Extended Factorial-Design Experiments

Beyond the six demonstration modules described in the main Results, the repository includes a set of more elaborate factorial-design experiments covering the same three paradigms (persona/survey, episodic memory, multimodal), each with explicit hypotheses grounded in prior literature. These are available under `examples/demonstration_suite/{02_association_memory,03_personality_survey,04_vlm_ta` in the code repository, alongside the raw and processed data already collected for each. Status reflects the state documented in each experiment’s own `README.md` as of this draft.

S1. Persona and survey experiments

Three sub-experiments testing how persona conditioning and cross-item memory affect model self-report, run against `smollm2:360m-instruct-fp16` (Ollama), temperature 0, $n = 5$ simulated participants per condition.

- **S1.1 — VVIQ-16.** Factors: Persona (WEIRD / non-WEIRD) $\times$ Memory (item-level SingleTurn / task-level Convo / trial-level Convo). 16-item vividness rating, 1–5 Likert. Status: data collected and analyzed ($n = 5$ simulated participants $\times$ 16 items $\times$ 2 personas $\times$ 3 memory conditions, 480 rated trials total). Persona conditioning had essentially no effect on mean vividness (WEIRD 3.69 ± 0.88 vs. non-WEIRD 3.70 ± 0.89 , pooled across memory conditions). Memory condition mattered a great deal, but not in a way that reflects genuine item-by-item imagery report: under the persona-conditioned SingleTurn (item-level, no cross-trial memory) condition the model returned a rating of exactly 4 on all 160 trials ($SD = 0$) regardless of item content — a response-degeneracy artifact, not a vividness judgment. The task-level Convo condition was nearly as invariant (4.06 ± 0.24). Only the trial-level Convo condition (paired encoding+rating within a single thread) produced ratings spanning the full 1–5 range (3.02 ± 1.28 , e.g. trial-level SD $5\times$ the item-level SD). Persona conditioning is the operative word: a companion, persona-free replication of the same task and model (`smollm2:360m-instruct-fp16`, `cogtype="no"`, no persona statement in the system prompt) does *not* show complete collapse under SingleTurn — ratings there are concentrated but not degenerate ($M = 3.81$, $SD = 0.66$, values restricted to 3–5), broadening under Convo memory ($M = 3.73$, $SD = 0.99$, values spanning 1–5) (?). Read together, the two results suggest persona conditioning and memory architecture interact rather than act independently: persona conditioning alone narrows the response range: adding a no-memory constraint on top of it collapses that range entirely, in a way memory manipulation alone does not. This is a small-model (`smollm2:360m-instruct-fp16`), demonstration-scale result, not a powered study, but it illustrates exactly the kind of artifact a manipulation-based design is built to surface: an aggregate or naturalistic-stimulus evaluation of this model would likely report a plausible-looking mean vividness score without revealing that the item-level,

persona-conditioned condition wasn't discriminating between items at all, or that persona and memory jointly caused it.

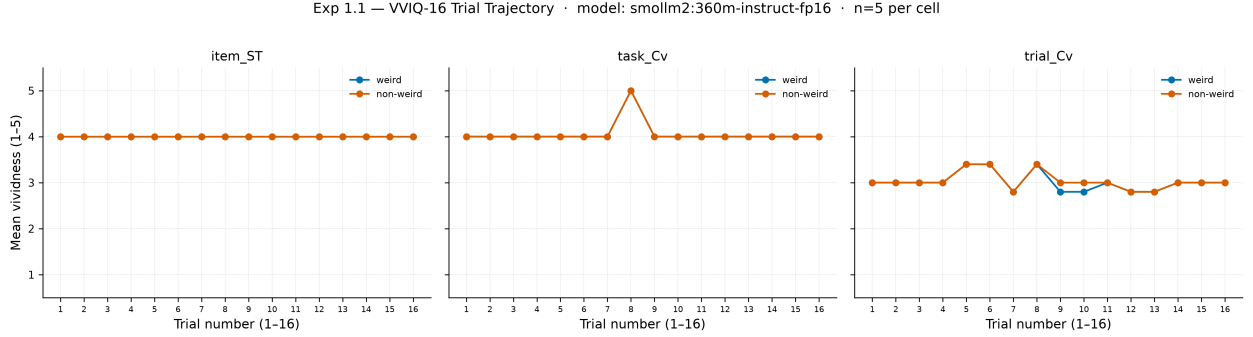


Figure 4: S1.1 (VVIQ-16): vividness ratings by trial, memory condition, and persona. `item_ST` (single-turn) is flat at rating 4 for every trial in both personas; `trial_Cv` shows genuine item-to-item variation.

Table 5: S1.1 (VVIQ-16): mean vividness rating ($\pm$ SD) by memory condition, pooled across personas ($n = 160$ trials/condition).

Memory condition	Mean vividness	SD
<code>item_ST</code> (single-turn)	4.00	0.00
<code>task_Cv</code> (episodic-chain)	4.06	0.24
<code>trial_Cv</code> (trial-chain)	3.02	1.28

- **S1.2a — BFI-44, 2-persona.** Factors: Persona (WEIRD / non-WEIRD+Jung) $\times$ Summary- k (1, 5) $\times$ Memory- k (4, 16), plus a SingleTurn control. Goal: measure how persona conditioning and conversation-summarization parameters affect BFI-44 self-report, benchmarked against OpenPsychometrics norms. Status: running.
- **S1.2b — BFI-44, 3-persona.** Adds a third persona condition (Zhang et al. individual descriptions, no cultural-axis framing) to the S1.2a design. Status: ready to run.

S2. Episodic memory experiments (PAL-50)

Two sub-experiments testing whether memory architecture affects paired-associate recall across five semantic-similarity levels (10 word pairs each, 0.0/0.25/0.50/0.75/1.0), same base model as S1.

- **S2.1 — With feedback.** Three memory architectures (single-turn baseline, task-level continuous memory, trial-level paired encoding+test) $\times$ correctness feedback (correct/incorrect + answer revealed) after each test trial. Status: running.
- **S2.2 — No feedback.** Same design, feedback withheld, for a feedback-contrast comparison against S2.1. Status: ready to run.

S3–S4. Multimodal experiments

S3 — Behavioral Profiling (BPT-10). 2×2 design: VLM type (OCR-rendered image vs. non-OCR text description) $\times$ feedback (correct/incorrect + answer, yes/no), 10 sequences of 10 study items plus one probe each, via OpenRouter (`nvidia/nemotron-nano-12b-v2-v1:free` and `mistralai/mistral-nemo-12b-instruct:free`). Measures planned: accuracy by serial position (lost-in-the-middle) and by probe-target modality (attentional residue). Status: ready to run.

S4 — VISER visual reasoning. Replication of a visual-structure counting-task manipulation (VISER, NeurIPS 2025 poster): OCR vs. non-OCR $\times$ grid-overlay/description present or absent, via `nvidia/nemotron-nano-12b-v2-v1:free`. Status: ready to run.

S5. Reproducibility: running psychscanner as a new user, and contributing a new task

The experiments above (S1–S4, and the two completed demonstrations in the main Results) are all built on the same public interface, runnable without an API key or GPU:

```
uv venv pyscan --python 3.11 && source pyscan/bin/activate
git clone https://github.com/saurabhr/psychscanner.git && cd psychscanner
uv pip install -e .

python -c "
from pathlib import Path
from psychscanner import ExpCardInit, ExpCard, ScannerModel, task_library, to_csv
task_path = task_library('rm_singleturn_demo', format='path', dirs='examples/tasks')
card = ExpCardInit(model='mock-llm', family='mock-llm', projectname='smoke_test',
                   proj_dir=Path.cwd()/'results', cogtype='no', nsim=1,
                   memory='SingleTurn', task_file=task_path)
scanner = ScannerModel(expcard=ExpCard(card))
to_csv(scanner, path=card.proj_dir)
"
```

`family="mock-llm"` needs no network access or credentials, so this same snippet is what we ask co-authors to run first when trying the framework rather than reading about it — swapping in a real `model/family` (see the Methods subsection on model providers) is the only change needed to move from the smoke test to an actual experiment. Any of S1–S4 or the two completed demonstrations can then be reproduced by pointing at that module’s own `run.py` in `examples/demonstration_suite/`, whose docstring states the exact model and expected outputs. Running an existing task card at cluster scale (Slurm, a self-hosted Ollama server, bulk data transfer) uses the same public `ExpCardInit` interface as the smoke test above — only `model`, `family`, and `parameters` change between deployment targets: (i) local, self-hosted via Ollama (the smoke test above, and every S1–S4 experiment); (ii) HiPerGator, a Slurm-scheduled Ollama server for runs too large for a laptop; (iii) UF Navigator, a hosted LLM gateway available to AIBHS-affiliated co-authors as an alternative to self-hosting; and (iv) hosted APIs directly (OpenAI, Anthropic, Groq, and others already used by two of the six demonstration modules). Everything else about a task card is unaffected by which of the four is used; cluster-scale onboarding materials (Slurm job scripts, credential setup) are maintained internally by the lab rather than in the public repository.

Composing a new, one-off experiment beyond the bundled task cards is ordinarily a matter of combining independently varying `ExpCardInit` fields rather than writing new framework code:

memory architecture (`memory`, `chain_type`), persona conditioning (`cogtype`, `persona_files`), response parsing (`parser`), trial-level feedback (`feedback_fn`), and conditional branching (`next_trial_fn`) are each a single field or a small subclass, crossed the same way S1.1's persona $\times$ memory design crosses two of them.

Contributing a new task card follows the same pattern as the bundled ones under `examples/tasks/` — `CONTRIBUTING.md` covers the PR workflow (lint/test via `nox`). One naming gotcha worth stating explicitly since it caused a real, silently-wrong-taskname bug in four of the bundled cards during this project's own development: a task card's internal `"taskname"` field must match its filename stem, or `task_library()` will resolve it under the wrong name with no error raised.

Availability

All task cards, run scripts, and any data already collected for S1–S4 are in the public repository at the paths above, under the same license as the main codebase.